22 September 1976
SUPERSEDING
MIL-R-43862(GL)
19 November 1973

MILITARY SPECIFICATION

RANGES, GAS, DOMESTIC (EXPORT)

This specification is approved for use by the Natick Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cabinet type, free-standing, flush-to-wall, domestic gas ranges with a top cooking section, an oven(s), and a broiler(s).

* 1.2 Classification. Gas ranges shall be of the following types as specified (see 6.2):

Type I - Single oven, 20 to 24 inches (508 to 610 mm) overall width
Type II - Double oven, 39 to 41 inches (991 to 1041 mm) overall width

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-R-11337 - Ranges, Packaging and Packing of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7310

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN GAS ASSOCIATION (A.G.A.)

Directory of Certified Appliances and Accessories

(Application for copies should be addressed to the American Gas Association, 8501 East Pleasant Valley Road, Cleveland, OH 44131.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

Z21.1 - Household Cooking Gas Appliances

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018 or the American Gas Association, 8501 East Pleasant Valley Road, Cleveland, OH 44131.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C 282 - Acid Resistance of Porcelain Enamels (Citric Acid Spot Test)
C 286 - Porcelain Enamel and Ceramic-Metal Systems
D 635 - Flammability of Self-Supporting Plastics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

GERMAN SPECIFICATION FOR GAS EQUIPMENT

DIN 3360 - Houseoven, Bakeoven, and Cooker for City Gas (Technical Terms, Construction, Parts, Performance and Control)

(Application for copies should be addressed to Beuth-Vertrieb Gmb H, Berlin W15 of Cologne, Germany.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.3).

3.2 Standard product. The ranges delivered under this specification shall be the manufacturer's commercial product, except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

3.3 Codes and standards. Gas ranges covered herein shall conform to the applicable requirements of ANSI Z21.1 and DIN 3360, as applicable.

3.3.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the contractor shall submit to the contracting officer, or his authorized representative, satisfactory evidence that the ranges he proposes to furnish under this specification meet the requirements of ANSI Z21.1 and DIN 3360, as applicable.

* 3.3.1.1 ANSI. Acceptable evidence of meeting the applicable requirements of ANSI Z21.1 shall be a reproduced copy of AGA Appliance Certificate; a listing of the range in the AGA Directory of Certified Appliances and Accessories; or a certified test report from a recognized, independent testing laboratory acceptable to the Government stating that the ranges have been tested and conform to the applicable requirements of ANSI Z21.1.

* 3.3.1.2 DIN. Acceptable evidence of compliance with provisions of DIN 3360 shall be a certificate stating that the ranges conform to the applicable requirements of DIN 3360.

* 3.4 Design. Ranges shall be white in color and shall be designed for floor mounting and for installation close against a vertical wall at the back and with sides completely exposed or with one side against a base cabinet and the other side against a wall or with both sides between base cabinets. The range shall be finished in accordance with 3.8. Unless otherwise specified (see 6.2), bright chrome trim strips shall be provided. The trim strip shall cover the crevice between the countertop of the wall cabinet and the top of the range side(s). Trim strip(s) shall be provided for installation along the depth of the range covering the crevice(s). The ranges shall be designed for satisfactory operation on manufactured gas, natural gas or the type of gas specified in the contract or purchase order. Type I ranges shall be provided with a recessed cooktop, one oven, one broiler and four top burners. Type II ranges shall be provided with two ovens, two broilers and four top burners. Each control valve shall be of the depress and turn type to prevent the control valve from accidental turn-on. Safety devices shall be provided to shut off the flow of gas when burner or pilot flames are accidentally extinguised as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

(1) Each top burner shall have a separate safety device, (2) The oven pilot(s) shall have a separate device that will shut off gas. Range design shall provide easy access to inspect, service, and adjust any component or equipment required for proper operation of the range. Provisions for permanently grounding the range shall be provided.

* 3.5 Dimensions. Ranges shall conform to the dimensional requirements specified in table I.

TABLE I. Dimensions inches (mm)

Measurement	Type I	Type II
Overall width	20-24 (508-610 mm)	40 ± 1 (1016 ± 25 mm)
Overall depth, including handles	25 ± 2 (635 ± 51 mm)	25 ± 2 (635 ± 51 mm)
Height 1/ from floor to cooking top	35 ± 1 (889 ± 25 mm)	35 ± 1 (889 ± 25 mm)
Oven interior:		
Width (min.)	16 (406 mm)	16 (406 mm) (each oven)
Depth (min.)	19 (483 mm)	20 (508 mm)
Height 2/ (min.)	9-1/2 (241 mm)	12 (305 mm) (main oven)

1/ Does not include raised edges, if provided.

2/ Height shall be from top of rack, when the rack is positioned in lowest rack support, to the oven interior top.

3.6 Construction. Ranges shall be constructed so parts are secure against distortion, displacement and warping and shall be supported to maintain a fixed relationship between parts. Parts not permanently secured shall be constructed so they cannot be incorrectly assembled or misaligned during replacement or cleaning.

* 3.6.1 Body. Side panels shall be one-piece extending from the range top to the base. When a toe space is provided, the space shall be minimum 2 inches (51 mm) deep by 2 inches (51 mm) high.

* 3.6.2 Broiler. The broiler(s) shall be a smokeless type (no smoke issue from flue) and shall be located either below or within the oven(s). Broilers shall be provided with a broiler pan and a removable grill. Pan shall be porcelain-enameled steel and grill shall be stainless steel, porcelain-enameled steel, or chrome-or nickel-plated steel. Doors when provided, for movable broilers shall be drop-type. Pull-out drawer type shall be mounted on nylon wheels without bearings, or ball or roller bearing glides, slides or wheels. Doors when provided, for stationary broilers shall be either a drop-type or slide-opening type. Drop shelf type doors shall remain fully open without securing. Broiler vent discharge openings shall be located to deflect flue gases away from rear and side walls.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

* 3.6.3 Oven(s). The oven vent shall deflect smoke and flue gases away from rear and side walls. The oven shall be equipped with a thermostat which will provide automatic temperature control of the oven. The oven thermostat shall be provided with a graduated scale to a high of not less than 500 degrees Fahrenheit (F) (260°C). The oven thermostat control shall have the bake and broil positions of temperature legibly and permanently marked. Oven construction shall prevent moisture and vapors from contacting oven insulation. Oven insulating material shall be of the nonsettling type (batt or blanket type) installed in accordance with the manufacturer's standard practice and shall prevent surface temperature from exceeding those specified in 3.6.10. Loose fill insulation shall not be used. Oven bottom may be removable for cleaning and burner access.

* 3.6.3.1 Oven racks. The baking oven racks shall be constructed of stainless steel, porcelain enameled steel, or chrome or nickel plated steel. Oven racks shall be provided with stops which will permit the rack to be withdrawn at least one-half its depth. When tested as specified in 4.6.4, racks extending up to 8 inches (203 mm) shall not deflect more than 1/8 inch (3 mm) and racks extending up to 11 inches (279 mm) shall not deflect more than 1/4 inch (6 mm). Racks extending other than indicated above shall be proportional to extension-deflection requirements indicated above.

* 3.6.3.2 Oven and broiler burners. The oven burner shall heat the oven to at least 320°F (160°C) for baking. Oven temperatures (220 to at least 500°F) (104-260°C) shall be thermostatically controlled. When one oven and broiler burner is provided, the burner shall be provided with a constant burning pilot. When two burners are provided, one for baking and one for broiling, individual constant burning pilots shall be provided for each burner. All burners shall be removable and replaceable.

3.6.3.3 Oven and broiler doors. Oven doors and broiler doors, as applicable, shall be provided with springs tensioned to hold the doors closed. The oven door shall be of the drop-shelf type provided with hinge springs to hold the door closed and when fully opened shall remain in that position without securing. Means shall be provided to adjust the tension on the oven door to maintain the door against the oven seal. The oven door and broiler door (when furnished) shall operate without binding or seizure when tested as specified in 4.6.7. Hinge springs for oven and broiler doors shall be located entirely outside the heated zone or protected from heat so spring life is not affected by heat. The surface temperature shall not exceed the requirements of 3.6.10 and, when tested as specified in 4.6.9, there shall be no evidence of charring, glowing, or flaming of the cheese cloth material.

3.6.4 Fuel system. The fuel system including burners and valves shall be designed to burn manufactured gas, natural gas or the type of gas specified (see 6.2). Gas valves shall be provided with a universal orifice, adjusted for the type of gas specified (see 6.2). Instructions for conversion to the other types of gas shall be furnished. Components of the fuel system shall be adequately protected and supported and shall be assembled so that there are no leaks in the fuel system (see 4.6.1). Gas supply lines incorporating

Source: https://assist.dla.mil -- Downloaded: 20260-21T22:39Z
Check the source to verify that this is the current version before use.

compression fittings shall have the nut torqued to a value of 55-60 inch pounds (6-7N.m) for 3/16 inch (5 mm) diameter tubing and for 3/8 inch (10 mm) diameter tubing the nut shall be torqued to a value of 125-130 inch pounds (14-15N.m). An adjustable gas pressure regulator shall be furnished with each range. The regulator shall be adjusted for the type of gas specified (see 6.2).

3.6.4.1 Burner ignition. Top, oven, and broiler burners, as applicable, shall be provided with pilot ignition. Pilots shall be constant burning type and shall ignite each burner automatically within 5 seconds after the gas is available at the burner (see 4.6.2). Ignition for the top, oven, and broiler burners, as required, shall be accomplished by depressing and turning the control knob to one of the heat positions. Separate pilots shall be provided for the oven burner and the broiler burner. Pilots shall be constructed so that normal drafts cannot extinguish pilot flame. Oven and broiler pilots shall automatically reignite burners in case of flame failure at all control knob heating positions. When each top burner is adjusted to low flame (1400 Btu/hr input) (41 W), (see 4.6.2), the top burner pilots shall automatically reignite top burners in case of top burner flame failure.

3.6.4.2 Automatic shut-off.

3.6.4.2.1 Top burners. Each top burner shall have an individual thermo-electric safety device that will shut-off the gas supply to the burner in not more than 150 seconds in case the pilot is extinguished under any circumstance (see 4.6.2). When the thermo-electric safety device is in the "OFF" position, there shall be a minimum clearance of 1/16 inch (2 mm) between the top of the thermo-electric safety device and the pilot safety valve push pull rod.

3.6.4.2.2 Oven and broiler. Each oven and broiler pilot shall have an individual automatic shut-off device (see 3.4), that will shut-off the gas supply to the pilot and burner it serves in not more than 90 seconds, in case the burner or pilot is extinguished under any circumstance (see 4.6.2).

3.6.4.3 Knobs and handles. Knobs and handles, including the knob bezel, shall be fabricated of plastic. Plastic material for burner valve control knobs and bezels, and door and drawer shall be molded and heat-resistant. The various positions shall be legibly and permanently marked on or immediately adjacent to each valve control knob. Controls or knobs shall not be installed on the cooking surface between burners. When tested as specified in 4.6.5.1 and 4.6.5.2, the plastic knobs shall be self-extinguishing and shall not break, chip, or crack. Knobs shall operate without binding and be secure when tested as specified in 4.6.1.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:39Z
Check the source to verify that this is the current version before use.

* 3.6.5 Top burner Btu (W) ratings. Each top burner shall have a normal Btu (W) hour input rating of 10,000 Btu (2930 W) for high flame and 1400 Btu (410 W) for low flame.

* 3.6.5.1 Burner design. Burner bodies and heads shall be made of brass, aluminum, cast iron, or steel or any combination thereof. Burner heads shall fit flush with the burner body when heads are two piece construction. The burner assembly components shall not show any space between the mating surfaces of the aluminum cap and bowl. The cast iron steel burner clearance between the mating surfaces of the cap and burner shall not exceed 0.010 inch (0.25 mm).

3.6.6 Grates. Top burners shall have one grate covering all burners or two grates covering two burners each, or individual grates covering each burner. Grates shall seat properly on the cook top. Grates shall be removable. Grate(s) shall be flat and shall not wobble when tested as specified in 4.6.3. The top working surfaces of the grates shall have no sharp edges, and shall be rounded as required for good enameling practice. The center of the grate shall not extend higher than the outside periphery of the grate.

3.6.7 Drip or aeration pan(s). The range shall have drip or aeration pan(s) located beneath the entire grate area to collect spillage. The pan(s) shall have a total capacity of not less than 16 ounces (0.45 kg) liquid measure (see 4.6.10).

3.6.8 Back guard. Each range shall be equipped with a back guard. The back guard shall be not less than 4 inches (102 mm) above the main top surface and shall extend the full width of the range top.

3.6.9 Insulation. Blanket or batt type, mineral or fiberglass insulation shall be used throughout the range. Insulation shall be installed to prevent settling and to limit surface temperatures on the exterior of the range in accordance with table II. There shall be no exposed insulation material within the burner box or any of the cooking compartments.

3.6.10 Surface temperatures. When tested as specified in 4.6.8, the surface temperature rise on the exterior surfaces of the ranges shall not exceed the values given in table II.

TABLE II. Surface temperatures

Component part	Maximum rise above ambient room temperature °F (°C)
Door panel	83° (28°C)
Door frames and front frames	83° (28°C)
Exterior sides and top	83° (28°C)
Handle and knobs:	
Metallic	54° (12°C)
Nonmetallic	90° (32°C)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

3.7 Accessories. Other accessories which are stock items furnished on current models being offered which are in excess of those specified herein, will be acceptable, except for the burner with an automatic temperature control. Removal of accessories that affect appearance or serviceability is not acceptable.

* 3.8 Finish. The ranges shall be white in color. Cooktops and oven door panels shall have a porcelain enamel finish. The range cooking top, drip or aeration pan, back guard, oven linings, broiler compartment, grates, side and front panels shall be finished in porcelain enamel or a suitable high temperature organic finish. Grate surfaces which are visible when in normal use position shall be coated with grey or black porcelain enamel. Top burners shall be finished in accordance with the manufacturer's standard practice. Exposed stainless steel surfaces, when applicable, high temperature organic finish shall be finished in accordance with the manufacturer's standard practice.

* 3.8.1 Porcelain enamel. The porcelain enamel shall be class A or better acid resistance when tested in accordance with ASTM C 282 and shall not be thicker than 12 mils (0.30 mm) (see 4.6.6). Porcelain enamel shall be glossy and free from discolorations, cracks, thin spots, holes, burn-off, chipping, copperhead, fish scaling, or enamel lumps, i.e., wrinkled or wavy or other imperfections, and shall not show bare metal at corners. Ground coat shall not show through cover coat on exterior surfaces, except as needed to give better breakage resistance at corners and around pierced holes. Porcelain enamel shall not have any evidence of crazing, crawling, specking or tearing which will affect appearance.

* 3.8.2 Metallic platings. Metallic plating shall not be missing and shall be complete. Components with metallic platings shall be free from blisters, flaking, peeling, pits and not smooth.

* 3.8.3 Unexposed surfaces. Surfaces of parts not exposed to view in normal usage shall be protected in such a manner that any oxidation which may occur will not interfere with its intended function. An oxide coating is acceptable for these unexposed surfaces. Side and front panels provided with organic finish shall be given a rust resistant primer coating on the unexposed surfaces.

* 3.9 Marking. Identification markings shall be permanently and legibly marked directly on the item or on an aluminum, brass, or stainless steel plate firmly affixed to the item. Marking shall be stamped, embossed, engraved or applied by photosensitive means. The plate thickness shall be not less than 0.012 inch (0.3 mm).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

* 3.9.1 Identification. Each range identification plate shall include the manufacturer's model and serial number, name, trade name or trademark, of such known character as to be readily identifiable with the manufacturer. In addition, such information as required by ANSI Z21.1 shall also appear on the nameplate.

3.10 Cleanability. The range design shall be such as to permit ease of cleaning all surfaces and cavities where grease and other forms of soil collect.

3.11 Workmanship. All components and assemblies of the range shall be free from dirt and other harmful extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings, molded parts, and stampings, if used, shall be free of sand, fins, pits, blow holes, and sprues. External surfaces shall be free from sharp edges and sharp corners. Component shall not be fractured, buckled, warped, loose, dented, punctured or malformed.

* 3.11.1 Metal fabrication. Metal used in the fabrication of the range shall be free from kinks. Forming and shearing shall not cause damage to the metal and shall be free of trim marks.

3.11.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be smooth and free of cracks, burn holes, undercuts or incompleted fusion. All scale and flux shall be removed from the finish weld area.

3.11.3 Fastening devices. Threaded fastener and rivet holes shall have burrs removed. Threaded fasteners shall not be missing or broken, cracked or stripped and shall be drawn tight. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole. Spring clips, when required, shall not be missing.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required, the range shall be examined for the defects listed in table III dimensions specified and tested as specified in 4.6 through 4.6.4, and 4.6.7 through 4.6.10. Failure to pass any test shall be cause for rejection of the first article. The presence of any defects listed in table III or any dimension not within specified requirements shall also be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

* 4.4.2 In-process inspection. Examination shall be made of the following operations to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to the items affected and the operations:

4.4.2.1 Visual examination of thermal insulation and finishes. The visual examination of thermal insulation and finishes shall be as follows:

(a) Thermal insulation. Installation of insulation shall be examined for compliance with 3.6.3 and 3.6.9.
(b) Finishes. Finish operations on unexposed surfaces shall be examined for compliance with 3.8.4.

4.4.2.2 In-process testing of the porcelain enamel. Tests shall be made to determine compliance with the requirements specified in 3.8.1 for porcelain enamel. Test method for thickness of porcelain enamel shall be as specified in 4.6.6. The lot shall be all like porcelain enameled components, for example, cooking top fronts, doors, or aeration bowls. The sample unit shall be one component, as applicable. Acid resistance of cooking tops shall be tested as specified in ASTM C 282. The inspection levels and acceptable quality levels (AQLs) shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

Requirement	Inspection level	AQL
Thickness of porcelain enamel	S-2	6.5
Acid resistance (cooktops only)	S-2	6.5

* 4.4.2.3 In-process testing of knobs and handles. Knobs and handles shall be tested for self-extinguishing and drop test as indicated in 4.6.5. The lot shall consist of all knobs and handles, as applicable, offered for inspection at one time. The sample unit shall be one knob or handle. The inspection level shall be S-1 with an AQL of 6.5 defects expressed in terms of defects per hundred units.

4.4.3 End item inspection. The lot shall consist of all ranges of one type offered for inspection at one time. The sample unit shall be one completely fabricated range.

* 4.4.3.1 Visual examination. Examination shall be made of the completely assembled range for the defects listed in table III. The inspection level shall be level II with AQLs of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE III. Classification of defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturer's standard product except for changes necessary to comply with specification requirements	X	
Finish (general)	Not type as specified	X	
	Metallic platings missing, incomplete, blistered, flaked, peeled, pitted, not uniform, or rough (when applicable)	X	
	Area of no film, runs, painted surfaces, area of rust		X
	Foreign matter imbedded in paint		X
	Components shall be free of fractures, buckles, warps, dents, or punctures	X	

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:39Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Porcelain enamel finish (sec 6.3) (for definitions of defects consult ASTM C 286)	Missing	X	
	Ground coat visible through cover coat, surface contains dark specks (when range is in the use position)	X	
	Panels of same color not uniform in depth of color or shade		X
	Chipped (exterior surface or surface not likely to contact food)		X
	Aeration bowls, burner trays, drip pans, oven racks, or any surface coming in contact with food: porcelain surface fractured, chipped, or blistered (include underside of broiler pan and top portion of oven interior liner)	X	
	Fish scaling	X	
	Copperheads accompanied by a pulling away or curling up of the enamel	X	
	Copperheads - depressed, (when range is in use position)		X
	Burn-off	X	
	Scale	X	
	Specking, crazing, hairlines, star marks, or pinholes		X
	Blisters exposing base metal	X	
	Tearing, curling, or crawling	X	
	Lump - one or more heavy lumps	X	
	Beads - any bead 1/32 inch (0.79 mm) or more	X	
	Scumming and lack of gloss	X	
Construction and workmanship (general)	Component part such as doors, racks, trays, drawers, broiler pans, aeration bowls, burner trays, or grates which do not fit properly	X	
	Cracked or fractured metal	X	
	Any component missing (except nuts, bolts, screws, or spring clips)	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (general) (cont'd)	Any burr, sliver, sharp edge, rough die, tool, trim, or grind marks	X	
	Any component fractured, buckled, warped, loose, kinked, dented, punctured, or malformed	X	
	Any nut, bolt, screw, or spring clip missing		X
Baking oven and broiling section	Broiler not mounted on type of bearing specified (if drawer type)	X	
	Oven vent blocked or obstructed	X	
Oven racks and broiler pan(s)	Oven racks not constructed of porcelain coated steel or chrome or nickel plated steel	X	
	Oven racks not provided with rack stops	X	
Oven and broiler door (as applicable)	Doors with spring tension not held closed	X	
	Will not remain fully open without securing	X	
	Means for adjusting tension not provided	X	
Cooking top	Not equipped with back guard	X	
	Back splash does not extend the full width of the cooking top	X	
Grates	Any tilt that may cause improper cooking performance, for example, range top, (including grates) not horizontal with base; cooking top warped, bowed, or otherwise not level	X	
	Center of grates higher than outside periphery (check visually using straight edge)	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Burners	Burners not removable and replaceable	X	
	Burner head (port section) does not fit flush to body (two or three piece burner only)	X	
	NOTE: A proper fit shall be determined by examining the burner assemblies against a lighted white (opaque) background with the mating surface horizontal to the eye and at a right angle to the background. The burner shall be rotated to provide examination of the entire periphery. The cap and bowl of aluminum burners shall not show any space between the mating surfaces when so examined. Cast iron and steel burners will be considered acceptable with a clearance not exceeding 0.010 inch (0.25 mm) between the mating surfaces of the cap and burner bowl.		
Burner trays and aeration bowls (including burner bowl type)	Warped or bent aeration pans		X
	Not removable (except when bowls are integral with top)	X	
Oven thermostat control (including tube and bulb as applicable)	Loose		X
	Marking less than 500°F (260°C)	X	
	Bake and broil positions missing or not legibly and permanently marked	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Knobs and handles	Not plastic	X	
Control knobs	Heat positions not legible or permanent	X	
	Controls installed between burners	X	
Accessories	Not as specified	X	
Welding	Missing	X	
	Not continuous, not smooth, not uniform, burn holes, cracks, fissures, incomplete fusion	X	
	Not finished to blend to adjacent surface	X	
	Not free from flux, scale, paint, or areas of other foreign material	X	
Threaded fasteners	Missing, stripped or cross-threaded, fractured	X	
	Loose		X
Rivets	Loose, not in full contact with members riveted	X	
Identification marking	Missing, incomplete, illegible		X

4.4.3.2 Dimensional examination. Examination shall be made of the gas ranges for compliance with dimensions specified. Any nonconformance to specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 6.5 defects expressed in terms of defects per hundred units.

4.4.3.3 End item testing. When a first article is not required, the first production unit shall be tested as specified in 4.6.7 through 4.6.10. Failure to pass the test shall be cause for rejection of the first article. Each range shall be tested in accordance with 4.6.1 and 4.6.2. Failure to pass the tests shall be cause for rejection of the individual range. Production ranges shall be tested as specified in 4.6.3, 4.6.4 and 4.6.10 using inspection level S-3 with an AQL of 4.0 defects expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

* 4.4.3.4 Inspection of preparation for delivery. Preservation-packaging, packing and marking shall be inspected in accordance with the quality assurance provisions of MIL-R-11337.

4.5 Certificate examination. Certificate of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of conformance with the specification requirements shall be examined and validated.

4.6 Tests.

4.6.1 Operational test. The operational test shall be conducted with the type of gas specified in the contract to within plus or minus 5 percent of the manufacturer's recommended hourly Btu (W) input rating and the gas pressure to the burners adjusted at normal pressure. The range shall be tested for leaks in the fuel system while the range is connected to a gas line (see 3.6.4). Burner control knobs shall be turned fully open and closed to check for looseness or binding (see 3.6.4.3). The door shall be checked to determine that it does not bind (see 3.6.3.3).

4.6.2 Burner ignition test. Pilots and automatic ignition and shut-off devices shall be adjusted and tested as specified in ANSI Z21.1 with the types of gas specified in the contract and as follows:

(a) The top burner control knobs shall be set at the "OFF" position and the pilots shall be ignited. One control knob shall be turned to the low position setting and the time required for the pilot to ignite the burner shall be noted. The knob shall then be turned to the "OFF" position. The knob shall then be turned to high position and the time required for the pilot to ignite the burner shall again be noted. The procedure shall be repeated for all top burners to determine conformance with 3.6.4.1. Each top burner in turn shall be ignited at low flame, the flame then extinguished and burner observed for reignition from the pilot to determine compliance with 3.6.4.1.

(b) With all top burner pilots turned off, depress and turn the control knob to the low position and ignite the burner. Extinguish the burner without moving the knob and check to determine gas is automatically cut off to that burner within the time specified in 3.6.4.2.1. Repeat the procedure with the knob set at full position and for all top burners.

(c) The oven and broiler pilots, automatic ignition and shut-off devices shall be tested for proper operation as specified in ANSI Z21.1 and to determine conformance with the time requirement specified in 3.6.4.2.2. Any nonconformance shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:39Z
Check the source to verify that this is the currentversion before use.

* 4.6.3 Grate test. A container with a 20 pound (9 kg) load shall be positioned on the grate to determine conformance to the grate wobble requirements of 3.6.6. Using a straight edge, check flatness requirements of 3.6.6. Inability of the grate to meet the requirements specified shall constitute failure of this test.

* 4.6.4 Oven rack test. An oven rack shall be withdrawn to the stop of any rack position and the following loads shall be distributed over the full area of the rack to determine that the rack does not deflect more than the amount specified in 3.6.3.1. Inability of the withdrawn oven rack to meet the load requirements specified shall constitute failure of this test.

Width of rack, inches (mm)	Load, pounds (kg)
14 to 18 (356 to 457 mm)	20 (9 kg)
18 and over (457 mm and over)	25 (11 kg)

4.6.5 Plastic tests.

4.6.5.1 Self-extinguishing test. The plastic knobs shall be tested as specified in ASTM D 635 to determine conformance to flammability requirements of 3.6.4.3. Inability to meet the requirements specified shall constitute failure of this test.

4.6.5.2 Drop test. The plastic gas cock knobs and control valve knobs on the sample range shall be dropped three times from a height of 4 feet (1.2 m) onto an asphalt or vinyl tile surface over a cement floor to determine compliance with strength requirements of 3.6.4.3. Any nonconformance to the requirements specified shall constitute failure of this test.

4.6.6 Porcelain enamel thickness. Porcelain enamel thickness shall be measured with a magnetic thickness gauge calibrated to an accuracy of plus or minus 0.001 inch (0.02 mm) to determine conformance to finish thickness requirements of applicable subparagraphs of 3.8.1. Any nonconformance to the requirements specified shall constitute failure of this test.

4.6.7 Oven door test.

4.6.7.1 Hinge test preparation. The mechanical apparatus required for the oven door cycling shall be designed and the door cycling conducted in such a manner that at the test's conclusion, no disassembly of the door or adjustment of any of its components will be necessary to conduct subsequent tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

* 4.6.7.2 Hinge test. With the oven thermostat set at 330 plus or minus 5°F (166 ± 3°C) above the ambient temperature, operate the oven for at least one hour to assure stabilization of the oven temperature. Without adjusting the oven controls, operate the oven door a total of 25,000 cycles closing to within 5 degrees of fully closed position and opening to within 5 degrees of the fully open position and return at a frequency of approximately 15 cycles per minute to determine compliance with 3.6.3.3. Upon completion of the 25,000 cycles, shut the oven down and without any adjustment, modification or disassembly of the oven door its hinges or any other door component, subject the oven door to a load of 25 pounds (11 kg) plus or minus 5 percent applied without impact to a 5 square inch (32 cm) plus or minus 10 percent area for 5 minutes at one outer corner of the fully open door. The other outer corner will be supported in its fully open position. After removing the load, the oven door shall be visually examined for proper fit and closure, at which time there shall be no warpage of the oven door or evidence of gaps between the oven face and the oven door when in the fully closed position, and the procedure repeated with the load applied to the other outer corner. The oven door shall again be visually examined for proper fit and closure. A distributed load of 50 pounds (23 kg) plus or minus 5 percent shall be applied without impact along a 4 inch (102 mm) plus or minus 10 percent strip midway between the hinged edge and the outer edge of the fully opened door for 5 minutes. Upon examination of the test described in 4.6.8 and 4.6.9, no breakage or permanent distortion shall result. Again without adjustment to the oven door or its components, place it in the closed position and conduct the surface temperature tests set forth in 4.6.8. Broiler door shall be tested to determine compliance with 3.6.3.3. Any noncompliance with the requirements specified shall constitute failure of this test.

* 4.6.8 Surface temperatures. Surface temperatures of the ranges shall be tested to determine conformance to the requirements of 3.6.10 and table I. The thermostat shall be set to maintain an oven temperature of 330 ± 5°F (166 ± 3°C) above room temperature. The ambient room temperature range shall be from 70°F (21°C) minimum to 85°F (29°C) maximum during the entire test. Temperatures shall be taken one hour after a constant oven temperature has been attained. Temperatures shall be determined by use of an indicating surface pyrometer or thermocouples securely fastened to the surface being measured. One reading shall be taken on the grip area of the door handles and knobs. The top, sides, and door panels (excluding glazed area) shall be divided into 16 equal areas with readings taken at the center of each area. Temperature measurements for frame areas shall be taken on a center line 1/2 inch (13 mm) in from the outside of the frame. Door frame measurements shall be taken at 6 inch (152 mm) increments; window frame measurements shall be taken at 3 inch (76 mm) increments. Glass window areas shall be divided into 4 equal sections and temperature measurements taken at the center of each section. Any noncompliance to the surface temperature requirements specified shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

* 4.6.9 Scorching test. Immediately following the test in 4.6.8, two test panels shall be secured against the range sides. The panels shall consist of 1/2 inch (13 mm) thick plywood and shall extend from the floor or surface on which the range rests to 1 inch (25 mm) above the cooktop surface and from the back edge of the range to 1 inch (25 mm) in front of the outside front surface of the range. The test panel surfaces placed against the range sides shall be covered with two layers of cheesecloth. The cheesecloth shall be untreated cotton fabric 36 inches (91.4 cm) wide, running 14-15 yards per pound (12.8-13.7 m per 0.45 kg) and having what is commercially referred to as a "count of 32 by 28". The panels shall remain secured to the range sides with the oven operating continuously at 330 ± 5°F (166 ± 3°C) above ambient temperatures for 8 hours, at which time they shall be removed and examined. Inability to meet the requirements of 3.6.3.3 shall constitute failure of the test.

4.6.10 Drip or aeration pan(s) capacity. The drip or aeration pan(s) capacity shall be tested to determine conformance with 3.6.7. Any nonconformance with the requirements specified shall constitute failure of this test.

5. PACKAGING

5.1 Levels A, B and C. Preservation-packaging and marking for each level of protection shall be in accordance with the applicable requirements of MIL-R-11337 (see 6.2).

6. NOTES

6.1 Intended use. The ranges are intended for use in domestic kitchens of civil and military quarters.

* 6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type range required (see 1.2).
(c) When a first article is required (see 3.1, 4.3 and 6.4).
(d) When bright chrome strips are not required (see 3.4).
(e) Type of gas to be burned (see 3.6.4).
(f) Selection of applicable levels of preservation-packaging and packing (see 5.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Armed Services Procurement Regulations ASPR 7-104.55. The first article should be a preproduction sample, initial production item, or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

6.4 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.5 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Army - GL

Review activity:

Army - MD

Preparing activity:

Army - GL

Project No. 7310-A484

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:39Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
US ARMY NATICK RESEARCH and DEVELOPMENT COMMAND
NATICK, MASSACHUSETTS 01760

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
DRXNM-TDG

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

Commander
US Army Natick Research and Development Command
ATTN: DRXNM-TDG
Natick, MA 01760

NATICK OP 1 Apr 76

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-R-43862A - RANGES, GAS, DOMESTIC (EXPORT)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:39Z
Check the source to verify that this is the current version before use.